DS Lab 2: Data Wrangling II

Based on Academic Performance Dataset

In [13]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
```

In [14]:
```
# Creating sample dataset
np.random.seed(42)  # ensures reproducible practical output
data = {
    'math_score': np.random.randint(55, 90, 30),
    'reading_score': np.random.randint(60, 100, 30),
    'writing_score': np.random.randint(55, 90, 30),
    'placement_score': np.random.randint(70, 105, 30),
    'club_join_year': np.random.randint(2018, 2022, 30)
}

df = pd.DataFrame(data)
df.head()
```

Out[14]:

	math_score	reading_score	writing_score	placement_score	club_join_year
0	83	63	77	102	2020
1	69	84	75	92	2020
2	62	73	70	93	2020
3	75	68	72	104	2019
4	73	85	78	91	2021

In [15]:
```
# Introduce missing values
df.loc[5, 'math_score'] = np.nan
df.loc[10, 'reading_score'] = np.nan
df.loc[15, 'writing_score'] = np.nan

df.iloc[25]
```

Out[15]:
```
math_score           57.0
reading_score        73.0
writing_score        61.0
placement_score      92.0
club_join_year     2020.0
Name: 25, dtype: float64
```

In [16]:
```
# Check missing values
df.isnull()
```

Out[16]:

	math_score	reading_score	writing_score	placement_score	club_join_year
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
5	True	False	False	False	False
6	False	False	False	False	False
7	False	False	False	False	False
8	False	False	False	False	False
9	False	False	False	False	False
10	False	True	False	False	False
11	False	False	False	False	False
12	False	False	False	False	False
13	False	False	False	False	False
14	False	False	False	False	False
15	False	False	True	False	False
16	False	False	False	False	False
17	False	False	False	False	False
18	False	False	False	False	False
19	False	False	False	False	False
20	False	False	False	False	False
21	False	False	False	False	False
22	False	False	False	False	False
23	False	False	False	False	False
24	False	False	False	False	False
25	False	False	False	False	False
26	False	False	False	False	False
27	False	False	False	False	False
28	False	False	False	False	False
29	False	False	False	False	False

In [17]:

```
# Fill missing values with mean (avoids chained-assignment warnings)
df['math_score'] = df['math_score'].fillna(df['math_score'].mean())
df['reading_score'] = df['reading_score'].fillna(df['reading_score'].mean())
```

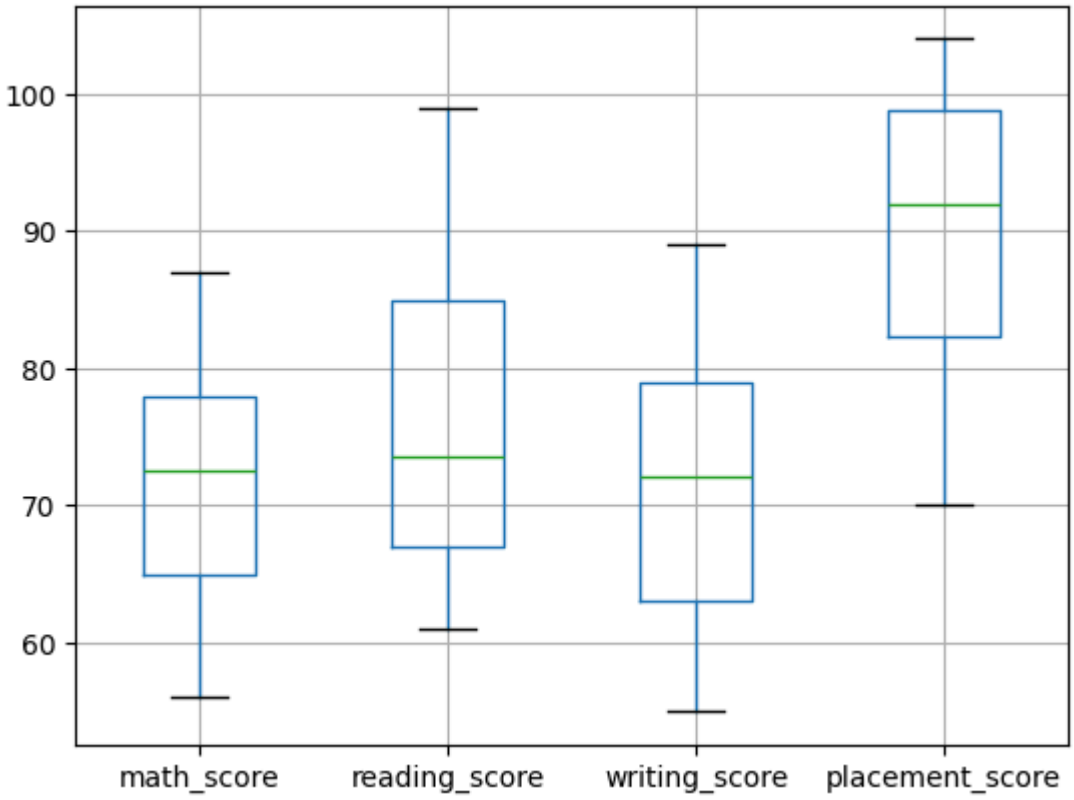
```
df['writing_score'] = df['writing_score'].fillna(df['writing_score'].mean())

df.head(20)
```

Out[17]:

	math_score	reading_score	writing_score	placement_score	club_join_year
0	83.000000	63.000000	77.000000	102	2020
1	69.000000	84.000000	75.000000	92	2020
2	62.000000	73.000000	70.000000	93	2020
3	75.000000	68.000000	72.000000	104	2019
4	73.000000	85.000000	78.000000	91	2021
5	71.310345	61.000000	80.000000	96	2019
6	65.000000	79.000000	79.000000	104	2019
7	65.000000	87.000000	83.000000	70	2018
8	78.000000	66.000000	69.000000	104	2019
9	78.000000	67.000000	55.000000	83	2018
10	57.000000	76.275862	79.000000	72	2018
11	76.000000	73.000000	61.000000	70	2019
12	56.000000	76.000000	63.000000	74	2021
13	78.000000	95.000000	78.000000	95	2021
14	84.000000	99.000000	55.000000	83	2021
15	56.000000	63.000000	72.275862	96	2021
16	75.000000	61.000000	78.000000	78	2021
17	87.000000	65.000000	65.000000	84	2019
18	66.000000	63.000000	71.000000	84	2019
19	76.000000	88.000000	62.000000	95	2020

In [18]:

```
# Detect outliers using boxplot
df.boxplot(['math_score','reading_score','writing_score','placement_score'])
plt.show()
```



In [19]:
```
# Z-score method
z = np.abs(stats.zscore(df[['math_score','reading_score','writing_score']]))
outliers = np.where(z > 3)
outliers
```

Out[19]:
```
(array([], dtype=int64), array([], dtype=int64))
```

In [20]:
```
# IQR method
Q1 = df['reading_score'].quantile(0.25)
Q3 = df['reading_score'].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df['reading_score'] < lower) | (df['reading_score'] > upper)]
outliers
```

Out[20]:

math_score	reading_score	writing_score	placement_score	club_join_year

In [21]:
```
# Replace outliers with median
median = df['reading_score'].median()
df['reading_score'] = np.where(
    (df['reading_score'] < lower) | (df['reading_score'] > upper),
    median,
    df['reading_score']
)

df.head(20)
```

Out[21]:

	math_score	reading_score	writing_score	placement_score	club_join_year
0	83.000000	63.000000	77.000000	102	2020
1	69.000000	84.000000	75.000000	92	2020
2	62.000000	73.000000	70.000000	93	2020
3	75.000000	68.000000	72.000000	104	2019
4	73.000000	85.000000	78.000000	91	2021
5	71.310345	61.000000	80.000000	96	2019
6	65.000000	79.000000	79.000000	104	2019
7	65.000000	87.000000	83.000000	70	2018
8	78.000000	66.000000	69.000000	104	2019
9	78.000000	67.000000	55.000000	83	2018
10	57.000000	76.275862	79.000000	72	2018
11	76.000000	73.000000	61.000000	70	2019
12	56.000000	76.000000	63.000000	74	2021
13	78.000000	95.000000	78.000000	95	2021
14	84.000000	99.000000	55.000000	83	2021
15	56.000000	63.000000	72.275862	96	2021
16	75.000000	61.000000	78.000000	78	2021
17	87.000000	65.000000	65.000000	84	2019
18	66.000000	63.000000	71.000000	84	2019
19	76.000000	88.000000	62.000000	95	2020

In [22]:

```
# Transform club_join_year to duration
current_year = 2026
df['duration'] = current_year - df['club_join_year']

df.head(20)
```

Out[22]:

	math_score	reading_score	writing_score	placement_score	club_join_year	duration
0	83.000000	63.000000	77.000000	102	2020	6
1	69.000000	84.000000	75.000000	92	2020	6
2	62.000000	73.000000	70.000000	93	2020	6
3	75.000000	68.000000	72.000000	104	2019	7
4	73.000000	85.000000	78.000000	91	2021	5
5	71.310345	61.000000	80.000000	96	2019	7
6	65.000000	79.000000	79.000000	104	2019	7
7	65.000000	87.000000	83.000000	70	2018	8
8	78.000000	66.000000	69.000000	104	2019	7
9	78.000000	67.000000	55.000000	83	2018	8
10	57.000000	76.275862	79.000000	72	2018	8
11	76.000000	73.000000	61.000000	70	2019	7
12	56.000000	76.000000	63.000000	74	2021	5
13	78.000000	95.000000	78.000000	95	2021	5
14	84.000000	99.000000	55.000000	83	2021	5
15	56.000000	63.000000	72.275862	96	2021	5
16	75.000000	61.000000	78.000000	78	2021	5
17	87.000000	65.000000	65.000000	84	2019	7
18	66.000000	63.000000	71.000000	84	2019	7
19	76.000000	88.000000	62.000000	95	2020	6

In [23]:

```
# Log transformation to reduce skewness
df['log_math'] = np.log10(df['math_score'])

df[['math_score','log_math']].head()
```

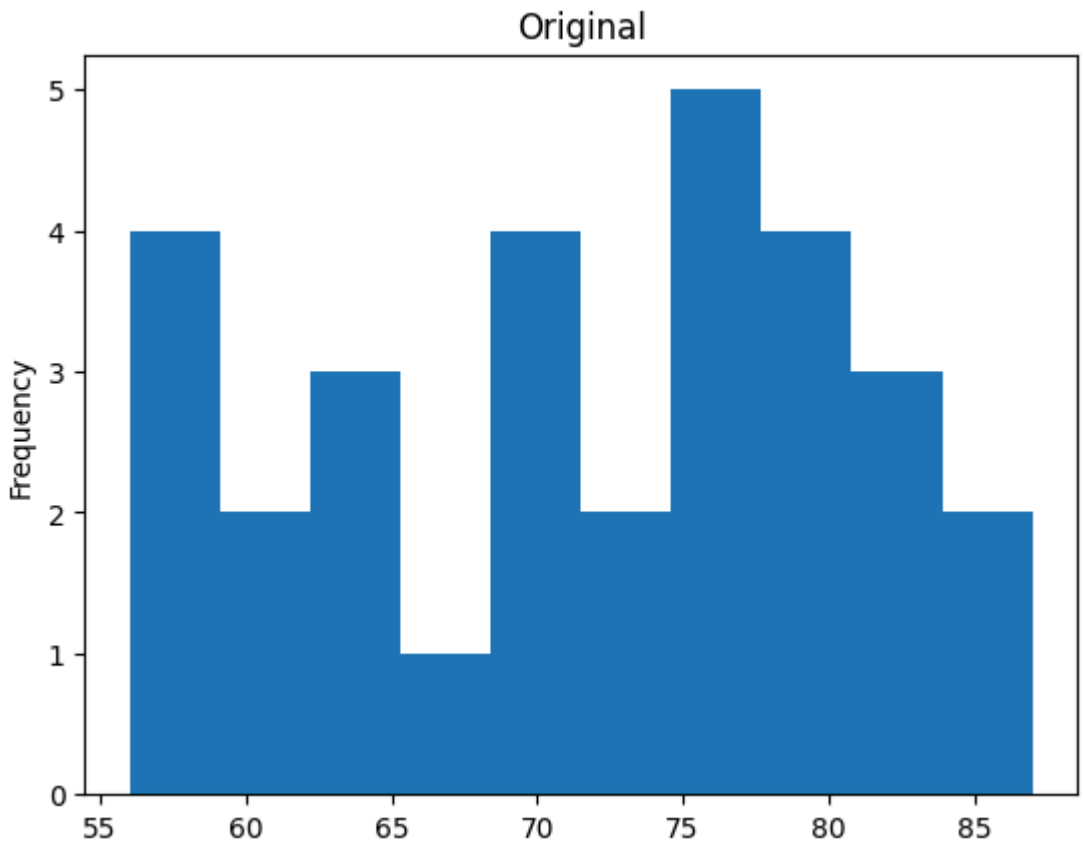
Out[23]:

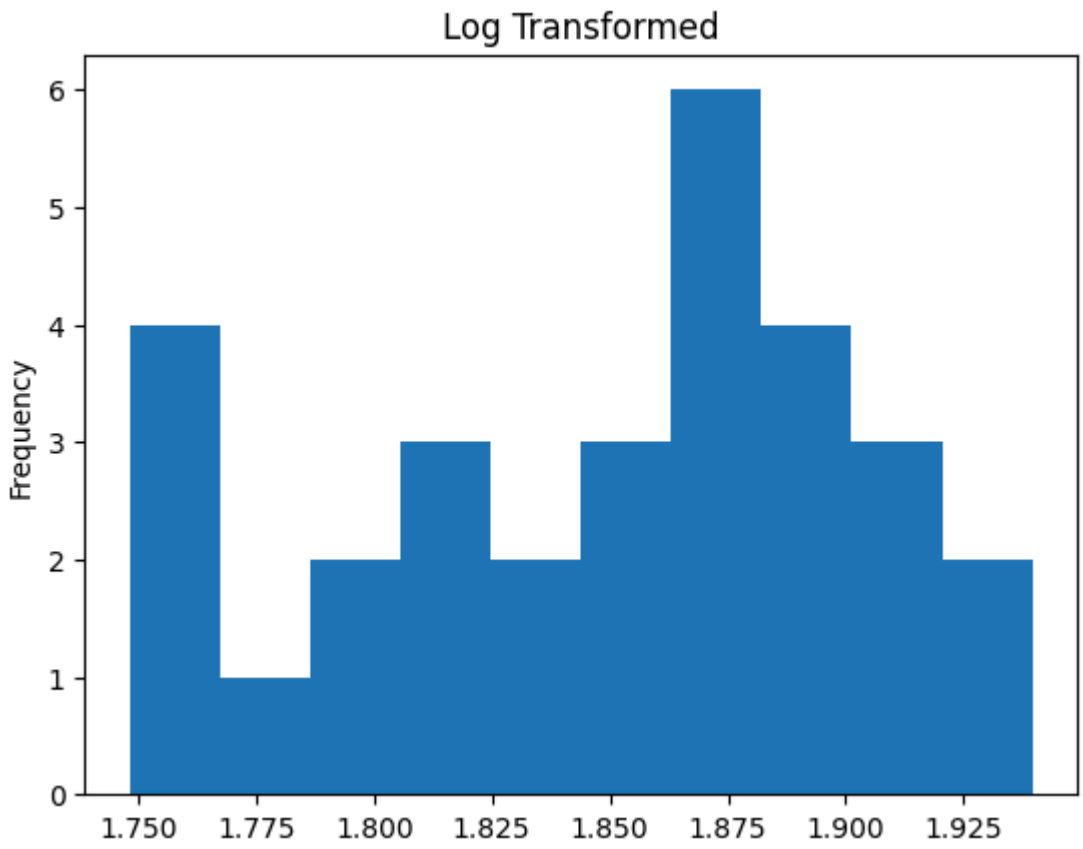
	math_score	log_math
0	83.0	1.919078
1	69.0	1.838849
2	62.0	1.792392
3	75.0	1.875061
4	73.0	1.863323

In [24]:

```
# Histogram comparison
df['math_score'].plot(kind='hist', title='Original')
plt.show()
```

```
df['log_math'].plot(kind='hist', title='Log Transformed')
plt.show()
```





In [ ]:

In [ ]: